

Fourteenth International Olympiad, 1972

**1972/1.**

Prove that from a set of ten distinct two-digit numbers (in the decimal system), it is possible to select two disjoint subsets whose members have the same sum.

**1972/2.**

Prove that if  $n \geq 4$ , every quadrilateral that can be inscribed in a circle can be dissected into  $n$  quadrilaterals each of which is inscribable in a circle.

**1972/3.**

Let  $m$  and  $n$  be arbitrary non-negative integers. Prove that

$$\frac{(2m)!(2n)!}{m!n!(m+n)!}$$

is an integer. ( $0! = 1$ .)

**1972/4.**

Find all solutions  $(x_1, x_2, x_3, x_4, x_5)$  of the system of inequalities

$$(x_1^2 - x_3x_5)(x_2^2 - x_3x_5) \leq 0$$

$$(x_2^2 - x_4x_1)(x_3^2 - x_4x_1) \leq 0$$

$$(x_3^2 - x_5x_2)(x_4^2 - x_5x_2) \leq 0$$

$$(x_4^2 - x_1x_3)(x_5^2 - x_1x_3) \leq 0$$

$$(x_5^2 - x_2x_4)(x_1^2 - x_2x_4) \leq 0$$

where  $x_1, x_2, x_3, x_4, x_5$  are positive real numbers.

**1972/5.**

Let  $f$  and  $g$  be real-valued functions defined for all real values of  $x$  and  $y$ , and satisfying the equation

$$f(x+y) + f(x-y) = 2f(x)g(y)$$

for all  $x, y$ . Prove that if  $f(x)$  is not identically zero, and if  $|f(x)| \leq 1$  for all  $x$ , then  $|g(y)| \leq 1$  for all  $y$ .

**1972/6.**

Given four distinct parallel planes, prove that there exists a regular tetrahedron with a vertex on each plane.